

PORTFOLIO BUILDING – B25CS0311

Activity 14: Hackathon Preparation

Team Name: PixelGuard

1. Activity Objective

The objective of this activity is to prepare for a hackathon by forming a team, selecting a suitable hackathon, and shaping a clear problem statement before the build phase.

The preparation focuses on understanding the problem, identifying a feasible solution, selecting a suitable technology stack, and defining responsibilities among team members.

2. Team Formation

Team: PixelGuard

Sl. No.	Team Member	Roll Number	Primary Role
1	T S HARSHAVARDHAN NAYAKA	R25EJ159	Project Development & Integration
2	Vighnesh P K KANHIRAKANDI	R25EJ174	Project Development & Testing
3	Terrance Paul S	R25EJ162	Project Development & Research
4	SHASHANK G NAIK	26017900366	UI, Documentation & Presentation

Team Approach

The responsibilities listed above represent the primary areas of contribution. The team will work collaboratively, and members may support one another across development, testing, research, interface work, documentation, and presentation whenever required.

3. Selected Hackathon

Semester End Examination; Hackathon – अभिनव (Abhinava)

Date: 07/10/2026

Time: 9:00 AM to 9:00 PM

Venue: SVB and SMVB Block

The team selected the Semester End Examination Hackathon – Abhinava for the hackathon preparation.

The hackathon requires a team of four students from the same section. Each team has to select one problem statement, and teams from the same section should not repeat the selected problem statement.

The hackathon has a duration of 12 hours, during which the team is expected to analyse the problem, identify a solution, design the system, and develop a working prototype.

4. Brainstorming of Problem Ideas

Before selecting the final problem, the team considered three different problem ideas.

Problem Idea	Feasibility	Potential Impact	Decision
Campus Lost & Found Board	High	High	Not Selected
Phishing URL Detection using Machine Learning	Medium–High	High	Not Selected
Image Tampering Detection: Real vs Edited	High	High	Selected

Reason for Selection

Image Tampering Detection was selected because the problem has a clear scope and can be demonstrated using image processing and machine-learning techniques. The proposed solution can produce a visible ELA heatmap and a classification result, making the core functionality suitable for demonstration.

5. Selected Problem Statement

Image Tampering Detection: Real vs Edited (Image Processing)

Digital images can be manipulated through techniques such as copy-paste, image splicing, and retouching, making it difficult to determine whether an image is authentic or edited.

The proposed solution is an image tampering detection tool that accepts an image as input and analyses it using **Error Level Analysis (ELA)** to highlight regions with inconsistent compression.

Features obtained from the image analysis will be used to train a simple machine-learning classifier to classify an image as real or potentially edited.

The system will provide an **ELA heatmap** and a **classification result** so that users can visually inspect the image and understand the detected inconsistencies.

6. Proposed Technology Stack

Area	Technology / Tool
Programming Language	Python
Image Processing	OpenCV, Pillow
Machine Learning	scikit-learn
Development Environment	Google Colab
Dataset	CASIA v2 or a small tampered-image dataset from Kaggle

7. Proposed Solution Flow

The proposed system will follow the following flow:

Input Image



Image Preprocessing



Error Level Analysis (ELA)



Feature Extraction



Machine-Learning Classifier



Prediction



ELA Heatmap + Classification Result

8. Proposed Features

The planned solution will include the following features:

1. **Image Input**
 - Accept an image for analysis.
 2. **Image Preprocessing**
 - Prepare the input image for further analysis.
 3. **Error Level Analysis**
 - Analyse compression inconsistencies in the image.
 - Generate an ELA representation/heatmap.
 4. **Feature Extraction**
 - Extract relevant features from the processed image.
 5. **Machine-Learning Classification**
 - Train a simple classifier to distinguish between real and potentially edited images.
 6. **Result Display**
 - Display the ELA heatmap.
 - Display the classification result.
 7. **Performance Evaluation**
 - Calculate classifier accuracy.
 - Generate a confusion matrix.
-

9. Expected Outputs

The expected outputs of the proposed system are:

- ELA heatmap of the uploaded image.
 - Classification result indicating whether the image is real or potentially edited.
 - Classifier accuracy.
 - Confusion matrix.
 - Sample image analysis results.
 - A working demonstration in which judges can upload an image and view the result.
-

10. Team Responsibility Plan

Team Member	Primary Responsibility
T S HARSHAVARDHAN NAYAKA	Project development and integration
Vighnesh P K KANHIRAKANDI	Project development and testing
Terrance Paul S	Project development and research
SHASHANK G NAIK	UI, documentation and presentation

The team will follow a collaborative approach during development. Members will assist each other whenever required for integration, debugging, testing, documentation, and presentation.

11. Preparation and Development Plan

Stage 1 – Understand and Plan

- Study the selected problem statement.
- Identify what the working solution should demonstrate.
- Break the problem into smaller tasks.
- Finalize the technology stack.
- Assign responsibilities among team members.

Stage 2 – Setup and Skeleton

- Set up the required Python environment.
- Verify required libraries.
- Obtain and organise the dataset.
- Create the initial project structure.

Stage 3 – Core Development

- Implement image preprocessing.
- Implement Error Level Analysis.
- Generate the ELA representation.
- Extract relevant features.
- Train the machine-learning classifier.

Stage 4 – Integration

- Connect the image input with the processing pipeline.
- Integrate ELA and feature extraction.
- Integrate the machine-learning classifier.
- Display the final result.

Stage 5 – Testing and Evaluation

- Test the system with multiple images.
- Check different input cases.
- Calculate accuracy.
- Generate the confusion matrix.
- Fix functional issues.

Stage 6 – Documentation and Presentation

- Prepare project documentation.
 - Prepare presentation slides.
 - Organise sample outputs and screenshots.
 - Prepare the live demonstration.
 - Keep a backup demonstration if required.
-

12. Expected Hackathon Deliverables

The planned project will aim to provide:

1. A working prototype/source code.
 2. README/documentation explaining:
 - Problem statement
 - Approach
 - Tools and libraries
 - Steps to run the project
 3. A 3–5 minute live demonstration.
 4. A backup screen-recorded demonstration.
 5. A 3–5 slide presentation containing:
 - Problem statement
 - Proposed solution
 - Technology stack
 - Challenges faced
 - Future scope
 6. Team details containing:
 - Team name
 - Member names
 - Roll numbers
 - Selected problem statement
 - Individual contributions
 7. Test/sample outputs and screenshots demonstrating the working solution.
-

13. Shared GitHub Workspace

The team will use the following GitHub repository as the shared workspace for the Activity 14 preparation:

GitHub Repository:

<https://github.com/harshavardhannayaka20072007-png/Activity-14-hackathon-preparation>

The repository contains the Activity 14 preparation materials, including the README and this report.

14. Preparation Conclusion

The team **PixelGuard** has been formed with four members, and the team has selected the **Image Tampering Detection: Real vs Edited** problem for the **Semester End Examination Hackathon – Abhinava**.

The preparation phase has covered team formation, problem brainstorming, feasibility and impact consideration, problem selection, problem understanding, proposed solution, technology stack, system flow, expected outputs, and team responsibilities.

The next phase will focus on implementing, integrating, testing, documenting, and demonstrating the proposed solution during the hackathon.

Activity 14 Completion Summary

Team Name: PixelGuard

Number of Team Members: 4

Selected Hackathon: Semester End Examination; Hackathon – अभिनव (Abhinava)

Hackathon Date: 07/10/2026

Selected Problem: Image Tampering Detection: Real vs Edited (Image Processing)

Primary Technologies: Python, OpenCV, Pillow, scikit-learn, Google Colab

Shared Workspace: GitHub Repository – Activity 14 Hackathon Preparation